

The empirical recurrence for OEIS A301883, recovered from its tail

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8 September 2026

Abstract

OEIS A301883 records “Empirical recurrence of order 52” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 715$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 3$, so the recurrence is proved for every $n \geq 55$. Its last coefficient is $c_{52} = -53934336 \neq 0$, so no further tail satisfies anything shorter; any order-52 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture that is not written down

OEIS A301883 is “Number of $n \times 7$ 0..1 arrays with every element equal to 0, 1 or 3 horizontally or antidiagonally adjacent elements, with upper left element zero.”. It has offset 1 and begins

21, 154, 823, 5463, 33853, 223869, 1454613, 9620565, ...

The entry states:

Empirical recurrence of order 52 (see link above)

As of the “Last modified” line on the live entry (Mar 28 2018 10:58:11, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 715$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 715$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 52: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 3$ is the first whose minimal order is exactly the stated 52.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 1430$ exact terms from $n = 3$ on, it returns order 52 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{52} c_i a(n-i),$$

c_1	1	c_{14}	9149335595	c_{27}	3100374816468	c_{40}	−35038755142440
c_2	124	c_{15}	1803433875	c_{28}	−89280035484614	c_{41}	1223944316376
c_3	−60	c_{16}	−66429153368	c_{29}	−3783566939570	c_{42}	11671458376044
c_4	−6860	c_{17}	−12401107053	c_{30}	138511315459321	c_{43}	−362042404032
c_5	1207	c_{18}	382135740011	c_{31}	2748242280547	c_{44}	−2763574879584
c_6	227174	c_{19}	64268091326	c_{32}	−171714500992171	c_{45}	63036863776
c_7	1852	c_{20}	−1756185205377	c_{33}	−44509036241	c_{46}	441458182240
c_8	−5078464	c_{21}	−254669519108	c_{34}	168803029388813	c_{47}	−5492888896
c_9	−581367	c_{22}	6482492581514	c_{35}	−2636522304340	c_{48}	−44007811968
c_{10}	81762420	c_{23}	774369335306	c_{36}	−130169204804013	c_{49}	150403584
c_{11}	13953906	c_{24}	−19276570541828	c_{37}	3529188544660	c_{50}	2407204800
c_{12}	−986311600	c_{25}	−1794395252812	c_{38}	77585464323388	c_{51}	2702592
c_{13}	−191682157	c_{26}	46221316767612	c_{39}	−2611603078672	c_{52}	−53934336

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 55$, and it is the recurrence OEIS A301883 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 3$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{52} - c_1 x^{51} - \dots - c_{52}$. Its constant term is $-c_{52} = 53934336$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 52 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 52, so $\deg q = 52$, and it is monic. It holds from some index t on. From $\max(t, 3)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 52, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 19 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 52, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -53934336 .

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-52 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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